Milestone 2 guidance

Preprocessing mixed hotel-booking data for Clustering

Hotel Booking Demand project · Mid-term technical progress / Milestone 2

Susana Nascimento (snt@fct.unl.pt)

NOVA FCT · Departamento de Informática · 2025/2026

Mixed Data Pre-processing

Advanced clustering methods still require a coherent representation

The data are mixed

Hotel bookings combine counts, dates, binary flags, nominal categories, high-cardinality fields, and outcome/post-event attributes.

The extensions are vector-space methods

FCM uses centroids and distances.
Ward hierarchical clustering is variance-based.
Spectral clustering builds a graph from pairwise similarities.

The main risk is methodological

Arbitrary category numbers, unscaled numeric variables, raw identifiers, or leakage variables can determine the clusters.

- **Use preprocessing to handle mixed variables**
- **Evaluate methods in the same representation**

  (otherwise comparisons are not meaningful)

Approach for the extensions

Keep standard Euclidean algorithms; solve mixed variables before calling the algorithm

Raw booking table → **Variable selection & representation design** → **Encoding transformation** → **Numerical scaling** → **R-EUCLID matrix** → **Standard algorithms**

Use the Euclidean distance but define carefully what each coordinate of the input space means.

Result: k-means, iK-means, FCM, spectral clustering and Ward hierarchical clustering, etc can be compared using the same documented Euclidean representation.

Project requirements: operational recommendations

Brief requirement	**Operational recommendation**
RQ2: representation	Name the final data matrix after variables transformation and standardization, e.g., R-EUCLID. State the metric sentence, e.g.: *Euclidean distance on the final transformed data matrix …*
Leakage control	Use outcome/post-event fields only after clusters are formed. Exclude $\mathrm{is\_canceled}$, $\mathrm{reservation\_status}$, $\mathrm{reservation\_status\_date}$ and other post-decision fields from clustering inputs.
High-cardinality variables	Drop raw agent/company identifiers. For country, use top-N or a minimum-frequency rule plus "Other", or justify exclusion.
Metric consistency	Run baseline and extension on the same data matrix unless the run is explicitly labelled as a representation / preprocessing comparison.
Scaling / ADR sensitivity	Minimum controlled variant: StandardScaler vs RobustScaler on numerical variables only. If ADR is used or considered, report noADR vs withADR as a separate representation comparison.
Robustness requirement	Rerun the same methods (distinct K's), seeds and metrics. Report whether quality, stability and booking-profile interpretation change.

Minimum: main representation + StandardScaler vs RobustScaler + with/without ADR if ADR is considered

Illustrate what students must state

The Data Matrix transformation defines what a clustering algorithm treats as similar bookings

Meaning

After preprocessing, each booking becomes a vector of columns: transformed numerical variables, binary variables and one-hot indicators. Euclidean distance is then computed on those columns.

$$d^2(i,\ell) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_\ell\|_2 = \sum_{j=1}^{p_{\text{num}}} (z_{ij} - z_{\ell j})^2 + \sum_{q=1}^{Q} \sum_{r=1}^{m_q} (o_{iqr} - o_{\ell qr})^2.$$

Therefore, encoding, scaling, capping, grouping rare categories and adding/removing ADR all change the distance.

Reporting Example

"We used representation R-EUCLID-standard-countryTop15-noADR. Numerical variables [lead_time_log, total_nights, party_size, …] were scaled with StandardScaler; categorical variables [meal, market_segment, …, country_top15+Other] were full one-hot encoded; one-hot columns stayed 0/1; ADR was excluded, etc ….. All clustering and internal indices used Euclidean distance in this final matrix."

Example: explain one pair of bookings before interpreting clusters

Feature block	Booking A	Booking B	Contribution to d²	Interpretation
z_total_nights	0.50	1.50	$(0.50-1.50)^2 = 1$	one scaled numerical difference
meal full one-hot	BB = (1,0,0)	HB = (0,1,0)	$(1-0)^2+(0-1)^2 = 2$	a nominal mismatch changes two dummy coordinates
ADR	excluded in noADR	excluded in noADR	0	include only in withADR representation

Common mistake to prevent: using raw integer codes such as month = 1, 2, …, 12 or room type = 1, 2, … creates artificial distances.

StandardScaler *vs* RobustScaler, and with/without ADR

Treat these as predefined sensitivity checks; keep the clustering protocol fixed

StandardScaler

$z = (x - mean) / range$

Recommended baseline for transformed numerical variables. Sensitive to extreme values, so it should be checked when heavy tails exist.

RobustScaler

$r = (x - median) / IQR$

Controlled variant for skewed or outlier-prone variables. It reduces the influence of extreme numerical values on Euclidean distance.

Important

Do not apply either scaler to one-hot columns. Keep dummy variables 0/1 unless a block-weighting rule is explicitly justified.

Recommended experiment rows to include in experiments.csv

Representation_id	Numerical scaling	ADR input?	Question tested	Report
R0-standard-noADR	StandardScaler	No	Main behavioural booking representation	metrics, stability, cluster profiles
R1-robust-noADR	RobustScaler	No	Are clusters driven by outliers / heavy tails?	same K-grid, seeds, metrics; compare ARI and profiles
R2-standard-withADR	same as R0 + scaled ADR	Yes	Do clusters become price / value segments?	required comparison if ADR enters any model

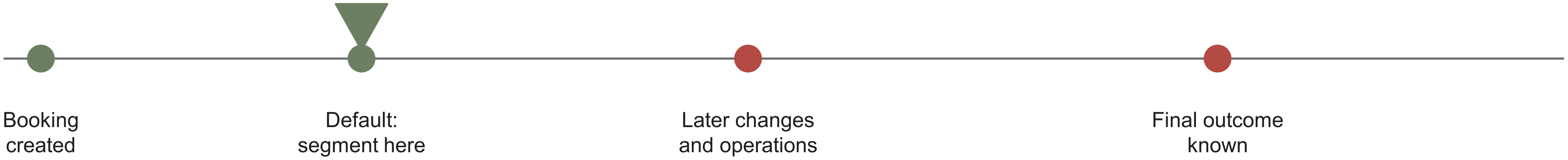
ADR rule: default behavioural segmentation = noADR.
If ADR is included, treat extreme values with a pre-specified rule, scale it as numerical, and explicitly compare withADR vs noADR.
If the interpretation changes, say that the representation now measures commercial-value similarity.

First decision: define the segmentation time

Segmentation / assignment time

The moment when the hotel would assign a booking to a segment. The clustering model can only use the booking snapshot available at that moment.

Recommended default: segment immediately after the initial booking is recorded.



Variables allowed under this default

lead_time; intended arrival period; reserved_room_type; stay length; party composition; meal; market_segment; distribution_channel; customer_type; deposit_type; previous-customer history; special requests if recorded at booking.

Not clustering inputs under this default

is_canceled; reservation_status; reservation_status_date; assigned_room_type; booking_changes; any field created only by later hotel operations or by the final outcome.

Eligibility test **Could the hotel know this exact value at the segmentation time, before later operations or the outcome?**

If the answer is “no”, keep the variable out of the clustering inputs and use it only for post-hoc profiling.

Example: deciding feature eligibility

Booking created on 10 Jan for a 15–18 July stay; default segmentation occurs immediately after creation

Variable in the dataset	Value / event in the example	Decision at 10 Jan	Reason
lead_time	186 days before arrival	INPUT	Known when booking is created. Transform/scale if used.
reserved_room_type	A	INPUT	Intended room type selected at booking; one-hot encode.
arrival month, nights, party size	July; 3 nights; 2 adults	INPUT	Known at booking; encode month cyclic/one-hot; scale numeric features.
ADR	€120 shown in the reservation	CONDITIONAL	Use only if price/value segmentation is part of RQ1 and price is known; otherwise report noADR and profile ADR later.
booking_changes	1 change on 20 Feb	EXCLUDE	It happens after the segmentation time, so it leaks later operations.
assigned_room_type	C assigned on 14 Jul	EXCLUDE	Produced by the hotel after the booking was created.
is_canceled / reservation_status	0 / Check-Out on 18 Jul	PROFILE ONLY	Outcome/post-event information; use later to describe clusters, not to form them.

Post-hoc profiling example: after clusters are formed, compare cancellation rate, reservation_status, assigned_room_type, and ADR summaries by cluster to interpret business meaning.

Feature Selection before modelling

Classify variables before any clustering method is fitted

1. Clustering inputs

Available at the chosen segmentation time and meaningful for similarity.

Examples: lead_time_log, total_nights, party_size, meal, market segment, deposit_type, reserved_room_type.

2. Post-clustering profiling

Not used to form clusters, but useful to describe them after clusters are formed.

Examples: is_canceled, reservation_status, cancellation rate by cluster, ADR summaries if not used as input.

3. Exclude or compress

Leakage, post-decision variables, raw identifiers, or categories with too many rare levels.

Examples: raw agent, raw company, unrestricted country levels, booking_changes at booking time.

Deliverable: include a feature-selection table with variable | role (input/profile/exclude) | reason | transformation.

One-hot encoding

Represent a nominal category by a block of binary indicator columns

Definition

For one categorical variable, create one 0/1 column for each retained category.
A booking has value 1 in its category column and 0 in the others.

meal category	BB	HB	SC
BB	1	0	0
HB	0	1	0
SC	0	0	1

Why not integer codes?

Coding BB=1, HB=2, SC=3 would invent an order and unequal distances between meal types. Nominal categories do not have that order.

Why keep the full block?

In clustering we do not drop a reference level. Full one-hot treats all category mismatches symmetrically in Euclidean space.

For **nominal variables** such as meal, market_segment, distribution_channel, country, room types, deposit_type and customer_type, integer category codes are not acceptable.

Recommended: full one-hot encoding after rare-category grouping.

Categorical encoding: operational aspects

1 **Audit levels**
Count categories and rare levels.

2 **Group rare levels**
Use top-N or a **minimum-frequency threshold**, then “Other”.

3 **Encode**
Apply full one-hot; keep all retained categories.

4 **Handle unknown categories**
UseOneHotEncoder(handle_unknown='ignore') so unseen levels do not break the pipeline.

5 **Do not rescale one-hot variables**
Keep 0/1 columns unless a stated block weight is used.

Variable type	Recommendation
country	example rule: keep top 15 countries + Other; or drop if geography is not central to the question
agent/company	drop raw IDs; if useful, replace by has_agent / has_company
meal, market, channel, room, deposit, cust. type	full one-hot after rare-category control
month/seasonality	one-hot arrival month or a cyclic sin/cos representation; never raw month number as a distance variable

The goal is not to retain every raw level. The goal is to build a defensible Euclidean representation in which distances reflect meaningful booking similarity.

Ordinal variables: first decide what is truly ordered

Order may be meaningful; numerical spacing is a modelling assumption

Discussion

An ordinal variable has ordered categories, such as low < medium < high.

The order is meaningful, but the gap between levels is not automatically known.

Main risk in Euclidean clustering

The encoding becomes the distance. Coding short=1, medium=2, long=3 assumes equal spacing.

Coding room_type A=1, B=2, C=3 invents an order that does not exist.

Recommendation

Add an “ordinal policy” row to the feature-selection table before fitting any clustering model.

One-hot encoding *vs* cumulative binary encoding

These encodings reflect different assumptions about the measurement scale.

One-hot encoding

Use for nominal categories, where no meaningful order exists.
Creates one binary column per category.
All category mismatches are equally distant.

Cumulative binary encoding

Use for ordinal categories, where the order is meaningful but equal spacing is not assumed.
Creates binary columns that accumulate the order information.
Adjacent categories are closer than distant categories.

Illustrative example: ordered stay-length band with categories short < medium < long

Encoding	Codes for short / medium / long	Distance implication	Interpretation
One-hot	(1,0,0) | (0,1,0) | (0,0,1)	d^2(short, medium)=2 d^2(short, long)=2	The encoding ignores order, so medium is not closer to short than long is.
Cumulative binary	(0,0) | (1,0) | (1,1)	d^2(short, medium)=1 d^2(short, long)=2	The encoding preserves order, so short is closer to medium than to long.

Key distinction: one-hot encoding ignores order, whereas cumulative binary encoding preserves order without imposing equal numerical intervals.

Recommended encoding for ordinal variables

First identify the measurement scale; then choose the representation

1 No real order (nominal) → one-hot

Use when the “order” is really a label or a business code.

Example: room type letters, meal type, customer type, deposit type.

Effect: all category mismatches have equal distance.

2 Order known, spacing uncertain → cumulative binary encoding, if needed

Use for low cardinality ordered categories

Example: short < medium < long stay.

Effect: adjacent levels are closer than distant levels without imposing equal numerical intervals.

3 Interval/ratio numerical variable → treat as numerical

Use when the feature is a genuine measured quantity or count.

Example: total requests, lead time, total nights.

Effect: distance assumes the numerical differences are meaningful after transformation and scaling.

Do not ordinal-code alphabetic room types, month numbers, country codes, market segments, or other nominal labels.

Do not present genuine counts or measured quantities as ordinal categories.

Decision rule for ordinal variables

Determine the scale first; choose the encoding only afterwards

Do not ordinal-code alphabetic room types, month numbers, country codes, market segments, or other nominal labels.

What is the measurement scale?

Nominal → one-hot.
Numerical (interval/ratio) → treat as numerical.
Ordinal → continue.

→

For an ordinal variable, is equal interval spacing defensible?

No → cumulative binary encoding or one-hot.

→

Could it dominate?

Check dimensionality, sparsity, scaling, and distance contributions.

→

Main or sensitivity?

Record representation_id and rerun metrics.

Practical interpretation

Nominal labels → one-hot.

Ordinal categories, spacing not defensible → cumulative binary encoding (or one-hot if there are very few levels).

Numerical interval/ratio variables → transform and scale as numerical variables.

Use rank/integer coding for an ordinal variable only if equal spacing between levels is substantively defensible.

Hotel data: treatment of "ordinal-looking" variables

Variable family	Examples	Recommended treatment
Measured counts / amounts	lead_time; total_nights; party_size; previous_*; total_of_special_requests; required_car_parking_spaces	Treat as numerical. Transform/cap if needed, then scale. Do not replace with arbitrary ordinal codes unless justified.
Created ordered bands	lead_time_band; stay_length_band; request_band	Optional. Prefer only for interpretability or sensitivity. Use cumulative binary or one-hot; compare against the original numerical version.
Cyclic time variables	arrival_date_month	Not ordinal in Euclidean distance. Use one-hot month or cyclic sin/cos representation.
Nominal labels with codes	reserved_room_type; meal; market_segment; distribution_channel; customer_type; deposit_type; country	Full one-hot after rare-category governance. Country: top-N/minimum-frequency + Other if used.
Leakage / post-event fields	is_canceled; reservation_status; reservation_status_date; assigned_room_type; booking_changes by default	Not clustering inputs under booking-time segmentation. Use only for post-hoc profiling or exclude.

Example: an ordered stay-length band

The same information can imply very different Euclidean distances

Setup

Suppose the team deliberately creates stay_length_band from total_nights: short = 1–2 nights; medium = 3–6 nights; long = 7+ nights.

Encoding	Codes for short/medium/long	Distances	What it assumes
Raw rank, unscaled	1 | 2 | 3	d^2(short, medium)=1 d^2(short, long)=4	Equal intervals and possibly too much influence if not scaled.
Scaled rank	0.0 | 0.5 | 1.0	d^2(short, medium)=0.25 d^2(short, long)=1.00	Linear spacing is meaningful after scaling.
Cumulative binary encoding	(0,0) | (1,0) | (1,1)	d^2(short, medium)=1 d^2(short, long)=2	Order is preserved; adjacent levels are closer.
One-hot	(1,0,0) | (0,1,0) | (0,0,1)	d^2(short, medium)=2 d^2(short, long)=2	Order is ignored; all mismatches are equal.

Recommended main representation: keep total_nights as a numerical feature and scale it with the numerical block.

How to Report ordinal-variable decisions

Add the decision to the feature-selection table, representation record, and robustness plan

Feature-selection table example

Feature	Decision	Encoding
arrival_date_month	input for seasonality	one-hot or sin/cos; not ordinal
reserved_room_type	input room category	one-hot; no alphabetic rank
total_of_special_requests	input count	cap at 3 if justified; scale
stay_length_band	sensitivity only	short < medium < long

Main representation: total_nights = weekend_nights + week_nights.
Numerical: transform/cap → StandardScaler.
Nominal: OneHotEncoder(handle_unknown="ignore").
Variant: stay_length_band → cumulative binary encoding.

Numerical variables: transform, then scale

Use transformations to reduce skewness; use scaling to make units comparable

Use logp transform

logp(x) = log(1 + x).

It is used for non-negative skewed variables. It keeps x = 0 at 0 and compresses very large values.

Examples in this dataset

lead_time; previous_cancellations; previous_bookings_not_canceled; days_in_waiting_list if used.

Use only when conceptually meaningful.

Then scale numerical variables

Baseline: StandardScaler.
Sensitivity: RobustScaler when outliers or heavy tails influence distances.

Variable family	Recommended operation	Reason
lead_time	logp → scale	right-skewed; raw values may dominate distance
nights / guests	derive total_nights (= week + weekend), weekend_share, party_size (= adults + children + babies)	avoid double-counting raw and derived variables
historical counts	cap if necessary; logp → scale	many zeros and a few large values
ADR, if used	cap/winsorise → RobustScaler	price/revenue proxy can dominate clusters

Note: Scale only the numerical variables. one-hot encoded categorical variables should remain as 0/1 columns.

Scaling and column weighting

Every scaling choice changes what Euclidean distance regards as similar

Numerical variables

Standardise transformed numerical variables so that units such as days, counts, and prices are comparable.

Recommended baseline: StandardScaler.

Binary and one-hot variables

Keep binary indicators and one-hot columns as 0/1. Do not variance-standardise these columns.

Reason: a rare category such as deposit_type=Refundable would become artificially influential if rescaled.

Optional block weighting

If one categorical variable should count approximately like one numerical variable, multiply its one-hot block by a stated constant. This is preprocessing, not a new clustering algorithm.

Scaling is part of the definition of similarity and must be reported as a modelling choice.

All internal indices must be computed using Euclidean distance in the same preprocessed representation used to fit the clustering model.

Euclidean distance after preprocessing

The encoded matrix defines the similarity used by the clustering algorithm.

$$d^2(i,\ell) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_\ell\|_2 = \sum_{j=1}^{p_{\text{num}}} (z_{ij} - z_{\ell j})^2 + \sum_{q=1}^{Q} \sum_{r=1}^{m_q} (o_{iqr} - o_{\ell qr})^2.$$

i, ℓ : two bookings; $\tilde{\mathbf{x}}_i$: final preprocessed vector for booking i;

z_{ij} : scaled value of numerical variable j; p_{num} : number of numerical variables;

$o_{iqr} \in \{0,1\}$: one-hot indicator for category r of categorical variable q;

Q : number of categorical variables; m_q : number of retained categories of variable q.

Numerical variables

If a scaled numerical feature differs by one unit, its squared-distance contribution is $1^2 = 1$.

Categorical variables

With full one-hot encoding, a mismatch such as BB = (1,0,0) and HB = (0,1,0) contributes 2.

Before interpreting clusters, students must document the exact input matrix used by the algorithm and state that distances were computed with ordinary Euclidean distance between rows of that matrix.

Illustrative example: two bookings

The same Euclidean formula can be read as a sum of contributions from each transformed feature block.

Transformed feature	Booking A	Booking B	Contribution to d^2
z_lead_time	0.80	−0.20	$(0.80+0.20)^2 = 1$
z_total_nights	0.50	1.50	$(0.50-1.50)^2 = 1$
meal	BB = (1,0,0)	HB = (0,1,0)	2
customer_type	Transient = (1,0)	Transient = (1,0)	0

One-hot encoding in the example

The category meal is represented by three indicators: BB, HB, SC. Booking A has BB and Booking B has HB.

Categorical contribution

Because the meal category is different, two binary coordinates change. Therefore the meal block contributes 2 to squared distance.

Distance calculation

$$d^2(A,B) = 1 + 1 + 2 + 0 = 4, \qquad d(A,B) = 2$$

Interpretation: the two scaled numerical variables contribute 2; the meal mismatch contributes 2; customer_type contributes 0 because it is identical for the two bookings.

Variable recommendations: main representation and ADR

Default booking-time segmentation: leakage-safe Euclidean representation; ADR included only when justified by the research question

Exclude from clustering inputs

is_canceled; reservation_status; reservation_status_date; assigned_room_type and booking_changes by default; raw agent/company identifiers; usually arrival year.

Encode or compress

country: top-N or minimum-frequency + Other, or exclude;
hotel: binary/one-hot if relevant;
meal, market_segment, distribution_channel, customer_type, deposit_type, reserved_room_type: full one-hot.

Engineer and scale

lead_time_log; total_nights; weekend_share; party_size; has_children; previous customer-history counts; parking; special requests.
Scale numerical variables only.

ADR: operational rule

Average Daily Rate is a price/revenue variable. Including it makes the distance partly measure commercial value, not only booking behaviour.

Case	Action	Required report
General booking segmentation	Exclude ADR from clustering inputs. Use ADR only for post-clustering profiling.	Use a no-ADR matrix. State that ADR was not used to form clusters.
Value or pricing segmentation	Create a separate ADR-inclusive matrix.	Justify availability at the index time and relevance to the research question.
If ADR is included	Check extreme values; cap only with a predefined rule; scale ADR with the other numerical variables.	Compare no-ADR vs ADR-inclusive results: metrics, stability, and profiles.

Exclude ADR for behavioural booking segments; include it only for explicit value/pricing segments.

ADR: two illustrative modelling choices

The use of ADR depends on the segmentation question and the stated index time

Example A — general booking profiles

RQ1: Which types of bookings have similar timing, stay design, party composition, channel, and request patterns?

Action: fit without ADR. Use ADR only after clustering to describe whether some profiles have higher/lower median price.

Example B — commercial-value profiles

RQ1: Are there booking segments that differ in price/value as well as booking behaviour?

Action: fit a separate ADR-inclusive representation. Preprocess ADR: check extremes, apply a predefined cap if justified, and scale it as a numerical variable.

Why scaling and the with/without ADR comparison are necessary?

Two otherwise similar bookings have ADR = €80 and ADR = €250.

If ADR is used raw: squared-distance contribution = $(250 - 80)^2 = 28{,}900$

If ADR is scaled ADR contribution = (scaled_ADR_B − scaled_ADR_A)²: e.g. a 1.5-unit scaled difference contributes $1.5^2 = 2.25$

If adding ADR changes the clusters substantially, the report must state that the representation now measures commercial-value similarity, not only booking-behaviour similarity.

Controlled preprocessing comparisons

Robustness should test plausible representation choices, not search indefinitely

Priority	Comparison	Question answered
Recommended minimum	StandardScaler vs RobustScaler	Are clusters driven by outliers or heavy-tailed numerical variables?
Strong if geography appears dominant	with country_topN vs without country	Are the clusters mainly geographic groups?
Required if ADR is included	with ADR vs without ADR	Are the clusters mainly price or revenue bands?

Report for each comparison: internal indices, stability measure such as ARI, and whether the booking-profile interpretation changes.

Example of a predefined robustness protocol

Define one main representation and one controlled variant before running the models

Run	Representation	What stays fixed / what to compare
R0 (main)	StandardScaler + countryTop15 + noADR	Same algorithms (k-means, iK-means, chosen extension); same K-grid; same seeds
R1 (variant)	RobustScaler + countryTop15 + noADR	Only preprocessing changes; all else stays fixed
Compare	Mean ± sd Silhouette/CH/DB; stability (e.g., ARI); runtime	Check whether cluster profiles and interpretation materially change
Decision	Conclusions are robust only if metrics and interpretation are similar across R0 and R1	State explicitly whether the conclusion changes, and why

Example: R0 = StandardScaler + countryTop15 + noADR; R1 = RobustScaler + countryTop15 + noADR. Same algorithms, same K-grid, same seeds, same metrics.

Distance check and representation record

Make preprocessing inspectable before interpreting any clustering result

Distance contribution check

For a random sample of booking pairs, estimate the average squared-distance contribution of each feature family: numerical, country, channel, stay design, preferences, history.

Representation record

Save representation_id, index time, included/excluded variables, transformations, rare-category rules, final dimensions, metric, sample rule, and seeds. Reuse representation_id in experiments.csv.

Record item	Example
representation_id	R-EUCLID-standard-countryTop15-noADR
metric sentence	Euclidean distance in the final encoded matrix
categorical rule	full one-hot after top-N/minimum-frequency grouping
scaling rule	transformed numerical variables scaled; one-hot columns not variance-standardised
sample rule	full data, or fixed sample size + seed for spectral/hierarchical

Can another team reconstruct exactly the matrix and logged experiment on which the algorithm computed distances?

Milestone 2: Expected output

Bring these artefacts as files/tables/screenshots that are already traceable to the repository

EDA and data-governance

- Dataset version/hash recorded
- Missingness and outlier summary
- Leakage exclusions justified

Main representation

- Feature-selection table
- ColumnTransformer/Pipeline sketch
- representation_id + metric sentence

Minimum modelling evidence

- *K*-means + iK-means
- Fixed K-grid (e.g., 3-8)
- Silhouette + CH/DB + runtime

Experiment Logging and sensitivity plan

- experiments.csv columns defined
- At least 5 seeds or resamples

Mid-term goal: students should already be able to show a reproducible representation, a fixed protocol, and first clustering evidence